

Inequalities

Solve • Graph • Flip

Symbols & Meanings



less than
"smaller than"



greater than
"larger than"



at most
"less than
or equal to"



at least
"greater than
or equal to"

Worked Example

Solve for x :

$$-2x < 6$$

- 1 Divide both sides by -2 .

$$\frac{-2x}{-2} > \frac{6}{-2}$$

- 2 Simplify.

$$x > -3$$

- 3 Flip the sign when you multiply or divide by a negative.

Graph on a Number Line

Open Circle ($<$ or $>$)

$$x > -3$$



Open circle means the endpoint is not included in the solution.

Closed Circle ($\leq$ or $\geq$)

$$x \geq -3$$



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Open circle = not included; closed circle = included.

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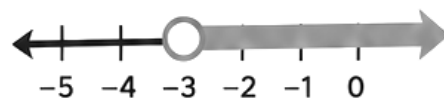
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Graph on a Number Line

Open Circle ($<$ or $>$)

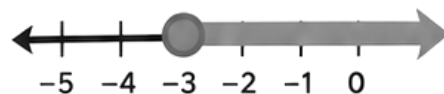
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